KARUPOTHULA GOUTHAM

Embedded Testing & Product Engineer

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PROFESSIONAL SUMMARY

Embedded testing and product engineer with about two years at Ripple Smart Metering, working in the gap between the R&D team and the production floor. Bring up new boards, test LoRaWAN gas electricity meters and SGLD and rework SMD boards by hand. Read and debug C/C++ firmware for ESP32 and STM32, using AI coding tools to speed up development and validating every build on real hardware. Design 2-layer PCBs in EasyEDA, so I see a layout from both the design bench and the assembly line.

TECHNICAL SKILLS

- **Hardware validation:** Board bring-up, power rail checks (3.3V, 5V), short-circuit and ground checks, active vs. deep-sleep current measurement with DMM, functional and regression testing.
- **Assembly and rework:** Fine-pitch SMD and through-hole soldering, IC replacement, solder bridge removal, jumper-wire fixes on prototypes, first-article visual inspection.
- **Microcontrollers and modules:** ESP32-WROOM-32D, ESP32-S3, ESP32-CAM, STM32WLE5, Ai-Thinker Ra-09H, GSM/cellular modules.
- **Protocols:** LoRaWAN (IN865, Class A and C, OTAA/ABP), UART, I2C, SPI, I2S, ESP-NOW, Wi-Fi, GSM, HTTP/REST webhooks.
- **PCB design:** EasyEDA: schematic capture, 2-layer layout, custom footprints, Gerber generation.
- **Software and tools:** Reading and debugging embedded C/C++, AI-assisted firmware development, PlatformIO, Arduino IDE, FreeRTOS (basics), Git, OrbiWise Network Server, serial monitor debugging.

PROFESSIONAL EXPERIENCE

Ripple Smart Metering, Hyderabad September 2024 – Present

Embedded Testing Engineer (R&D to Production)

- **R&D to production link:** Read new schematics and layouts from the design team, turn them into step-by-step assembly and test checklists and train the assembly technicians on the test procedures.
- **Pilot batch troubleshooting:** Inspect first-article runs for dry solder joints, solder bridges, reversed capacitors and diodes and misplaced parts. Hand-solder fine-pitch SMD ICs, replace damaged parts and add jumper-wire fixes to recover faulty boards and keep the schedule moving.
- **DFM feedback:** Keep track of defects that keep coming back from the line and pass them to R&D with specific suggestions on footprint size, component spacing and test pad locations for the next revision.
- **Bring-up and power profiling:** Check 3.3V and 5V rails for shorts before first power-on on new smart meter prototypes, confirm rail stability under load and measure deep-sleep current in the microamp range to verify battery-life targets for gas and water meters.
- **LoRaWAN meter validation:** Test smart gas and electricity meters on STM32WLE5 and Ra-09H modules on the IN865 band: OTAA/ABP joins, spreading factor adaptation, payload format, packet delivery and RSSI/SNR, all checked on the OrbiWise Network Server.

- **Metrology testing at FCRI, Kerala:** Took part in flow repeatability and pulse accuracy trials on DN25 ultrasonic water meters at the Fluid Control Research Institute, recording results at each flow rate against utility tolerances.
- **Firmware QA:** Develop C++ test routines in PlatformIO (with AI assistance) to stress UART, SPI and I2C buses. Log bugs with serial traces and steps to reproduce and re-test fixes with the firmware engineers to catch regressions.
- **Internal project, Voice Intercom and Keka Notification Terminal:** Designed and built a multi-node wireless intercom on ESP32-WROOM-32D with an INMP441 microphone and MAX98357A amplifier. Defined the system behaviour, developed the firmware with AI-assisted coding, then tested and debugged it on hardware. Tuned I2S DMA buffers for low-latency UDP voice over Wi-Fi, stopped feedback squeal by muting the local mic during transmit and linked Keka HRMS webhooks to play chimes at shift punch-in and punch-out, plus scheduled tea and lunch announcements.

PROJECTS

Smart Gas Leak Detector with Custom PCB: EasyEDA, ESP32-WROOM-32D, MQ-5, PlatformIO (C++)

- Designed a 2-layer PCB that keeps the hot MQ-5 sensor on the opposite side from the ESP32 and the onboard SMA antenna trace, with separate ground, to avoid thermal drift on the analog reading and RF interference.
- Developed non-blocking C++ firmware (AI-assisted, tested on hardware) with captive-portal Wi-Fi setup, an SSD1306 OLED, a buzzer alarm and automatic alerts over Telegram Bot API and SMTP email when the threshold is crossed.

Autonomous Astronomical Street Light Controller: ESP32-WROOM-32D, GSM module, C++

- Replaced LDR sensors, which fail when dirty or covered, with an astronomical algorithm on the ESP32 (AI-assisted code, validated on hardware) that works out sunrise and sunset daily from the location coordinates.
- Switched commercial lighting banks through industrial multi-pole contactors, with an on-unit push button so field staff can override on or off without a network.
- Built a local mDNS web dashboard over Wi-Fi with automatic fallback to GSM for remote control and status when Wi-Fi drops.

EDUCATION

B.Tech, Electronics & Communication Engineering Graduated April 2024

St. Martin's Engineering College, Hyderabad.